

Smart Storage System – Complete Schematic Diagram

(Environmental Monitoring + RFID Access Control + Alert System)

5V External Supply
(5V 2A Power Bank /
Regulated Supply)

5V (to modules)
GND (to modules)

DHT11
(Temperature & Humidity)

VCC → 5V
DATA → D2
GND → GND

Rain Sensor Module
(Rain / Water Detection)

VCC → 5V
AO → A0
GND → GND

MQ-2 Gas Sensor
(Gas-level Change Detection)

VCC → 5V
AO → A1
GND → GND

Ultrasonic Sensor HC-SR04
(Distance Detection)

VCC → 5V
TRIG → D6
ECHO → D7
GND → GND

RFID Card / Tag
(Authorized Access)
(Wireless input only)

RC522 RFID Reader

VCC → 3.3V ONLY
(NOT 5V)
GND → GND
SDA/SS → D10
MOSI → D11
MISO → D12
SCK → D13
RST → D8

16x2 LCD (I2C)

VCC → 5V
GND → GND
SDA → A4
SCL → A5

Servo Motor
(Door Lock / Open-Close)

Signal → D9

5V External Supply
(for Servo only)

VCC → 5V (external supply)
GND → Common GND

Buzzer

+ → A2
- → GND

HC-05 Bluetooth

VCC → 5V
GND → GND
TXD → D4 (Arduino RX)
RXD → D5 (Arduino TX)

Use a voltage divider / level shifting
for Arduino D5 → HC-05 RXD
(3.3V logic input).

ALL GND COMMON

Wire Color Legend

- 5V (Power)
- 3.3V (Power)
- GND (Ground)
- Signal (D2, D4, D5, etc.)
- Analog (A0, A1)
- I2C (SDA, SCL)

Pin Summary (Arduino UNO)

Component	Arduino Pin
DHT11 (Data)	D2
Rain Sensor (AO)	A0
MQ-2 (AO)	A1
Ultrasonic TRIG	D6
Ultrasonic ECHO	D7

Component	Arduino Pin
RFID RST	D8
RFID SDA/SS	D10
RFID MOSI	D11
RFID MISO	D12
RFID SCK	D13

Component	Arduino Pin
Servo Signal	D9
Buzzer	A2
LCD SDA	A4
LCD SCL	A5
HC-05 TXD (Arduino RX)	D4
HC-05 RXD (Arduino TX)	D5

Important Notes

- RC522 VCC → 3.3V ONLY (NOT 5V).
- Use a voltage divider / level shifting for Arduino D5 → HC-05 RXD (3.3V logic).
- Servo motor powered from external 5V supply (not from Arduino 5V pin).
- All GND must be common.